

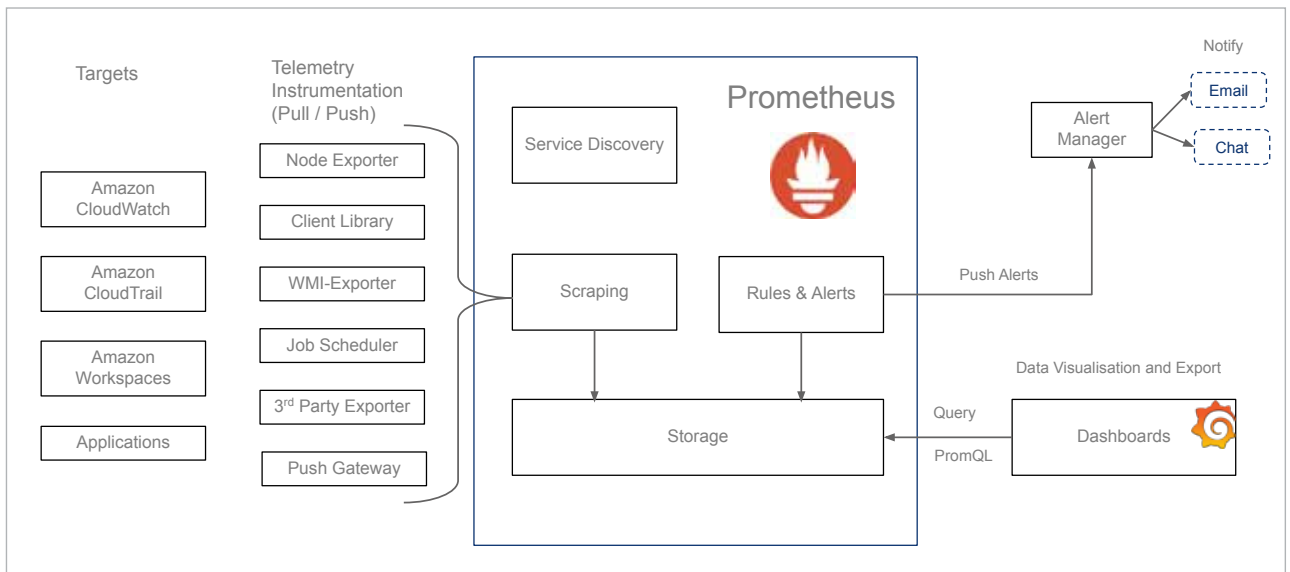


Figure 3: Open source tools for workspace observability

cloud monitoring where enterprises have to look for other external tools. There are a few popular open source tools that support these use cases as well and are a better choice from a cost point of view.

Prometheus is a popular open source tool used for event monitoring and alerting. It records real-time metrics using http pull from targets. It provides flexible queries and real-time alerting. Prometheus can be integrated with the cloud workspaces ecosystem. Figure 3 illustrates a sample integration of Prometheus with AWS Workspaces. Exporter jobs and services can be instrumented to pull metrics from workspace targets. Platform service metrics can be pulled from Cloud Watch and Cloud Trial services.

Grafana is a popular open source dashboard and visualisation tool with support for Prometheus as a data source. Granular dashboards can be prepared to monitor regional and departmental task based workspaces. Alerts can be configured in Prometheus by specifying a condition that needs to be maintained for a certain duration in order for an alert to trigger. When alerts are triggered, they are forwarded to the alert manager

service. Alert managers can include logic to forward these alerts to email, Slack, teams, or notification services such as PagerDuty.

Other popular open source tools include Istio and Jaeger. Istio supports different types of telemetry (metrics, distributed traces, and access logs) in order to provide overall observability. Jaeger is a tracing system that supports the open tracing standard. It enables distributing tracing, root cause analysis and service dependency analysis.

Beyond observability

Native observability patterns provide visibility based on external factors. To get visibility into internal conditions, we will still need custom third-party tools. These tools deploy custom

agents to gather specific data and provide detailed visualisation patterns. For example, to observe performance of applications running inside a desktop, we need to deploy additional application-specific custom agents.

In addition, we should also look at applying chaos engineering principles, which help to attain consistent workspace service reliability by hardening services against failures. **END** 🐧

Disclaimer

The views expressed in this article are that of the authors and Wipro does not subscribe to their substance, veracity or truthfulness.

By: Narasimha Sekhar Kakaraparthi and Radhakrishna Singuru

Narasimha Sekhar Kakaraparthi has 27 years of experience in product development for infrastructure management, cloud, as well as telecommunication, manufacturing and technical delivery. He is currently working as principal architect for virtual desktops and end user services at Wipro Technologies. He is a senior member of Wipro's eminent architects group called Distinguished Member of Technical Staff (DMTS).

Radhakrishna Singuru has more than 25 years of industrial experience in product and system software development spanning multiple industry domains. His areas of interest include cloud and virtualisation technologies, security engineering, blockchain, confidential computing, quantum computing, AI/ML, scalable platforms, network switching software, etc. He is currently working as chief architect, Cloud Platform Products and Security Engineering Practice, ER&D at Wipro Technologies. He is also a senior member of DMTS.